

Internship report

Development of a COMSOL Finite Element Breast Model for Deformation-Based Early Warning Scan Imaging



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1 Abstract

2 Introduction

Breast cancer is the most commonly diagnosed cancer among women worldwide, with more than 2.3 million new cases reported annually [1]. Early detection is essential for improving treatment options and patient outcome. X-ray mammography is currently the standard method for population-based breast cancer screening, but it also has relevant limitations. In particular, mammographic sensitivity is reduced in women with dense breast tissue, where the contrast between lesions and surrounding fibroglandular tissue can be limited [2, 3]. In addition, mammography uses ionizing radiation and requires breast compression, which motivates continued research into complementary, non-invasive and radiation-free screening approaches.

2.1 Background

One possible direction for radiation-free breast cancer detection is to use the mechanical contrast between healthy and pathological tissue. Malignant breast tumours are generally stiffer than surrounding adipose and fibroglandular tissue, although reported stiffness values vary with measurement technique, loading condition and tissue type [4–6]. This mechanical contrast forms the basis for elastography-based imaging techniques, in which tissue deformation is used to infer differences in internal mechanical properties [7–9].

This project was performed in collaboration with Early Warning Scan (EWS), which is developing a non-invasive screening concept based on optical and biomechanical measurements of breast deformation [10]. The underlying principle is related to Digital Image Elasto-Tomography (DIET), where the breast is mechanically excited and the resulting surface motion is captured using optical measurements. Differences in internal stiffness may then affect the measured surface displacement field and dynamic response [11–13]. Previous phantom and modelling studies have shown that finite element (FE) models can support the development of DIET-like systems by generating controlled deformation data and by testing the effect of stiff inclusions under known conditions [14, 15].

Finite element method (FEM) modelling is widely used to study breast biomechanics because it allows geometry, material properties, boundary conditions and loading conditions to be controlled independently. Early breast FEM studies already demonstrated the usefulness of three-dimensional biomechanical models for simulating breast deformation and image-related applications [16, 17]. More recent work has moved toward patient-specific and multi-component models that include separate anatomical regions such as adipose tissue, fibroglandular tissue, skin, muscle and support structures, each with distinct mechanical properties [16–19]. Such models can be used to study breast motion, compression, gravity-induced deformation, image registration and the effect of internal tissue composition on the external deformation response.

However, breast modelling remains challenging. Breast tissue is heterogeneous, nonlinear and individual-specific, and its mechanical properties vary substantially between studies, measurement protocols and anatomical regions [5, 6, 20, 21]. Anatomical features such as the posterior chestwall interface, fibroglandular distribution, skin layer and Cooper-ligament support can all influence the simulated deformation response. At the same time, adding anatomical detail increases model complexity, solver cost and the risk of poorly controlled comparisons. For an EWS-oriented modelling workflow, it is therefore important not only to make the model more realistic, but also to keep the development process reproducible and interpretable.

2.2 Objectives

The aim of this internship project was to develop a reproducible COMSOL-based FEM pipeline for a parametrised breast model and to use this pipeline to investigate how selected anatomical and mechanical modelling choices influence the simulated deformation response. The project started from the objective of improving the realism of an existing breast model, with an initial emphasis on geometry, posterior chestwall support and internal tissue representation. The focus was not to create a final patient-specific diagnostic model, but to establish a controlled model-development workflow in which model variants can be generated, solved and compared systematically.

The central research question of this project was:

How do controlled changes in chestwall geometry, glandular tissue distribution, outer breast shape, support representation, material settings and tumor inclusion affect the simulated displacement and stress response of a heterogeneous finite element breast model?

To address this question, the model was developed in a staged manner. The first stages focused on establishing a stable dynamic baseline and improving the anatomical description of the model through the posterior chestwall geometry and glandular tissue distribution. Subsequent stages introduced matched controls for outer-shape asymmetry and Cooper-ligament support, followed by material and skin-layer sensitivity cases and an initial tumor screening route. This staged structure was chosen to make the effect of individual modelling assumptions easier to interpret.

COMSOL Multiphysics was used as the intended modelling environment for this project, but an important part of the work was to determine whether a sufficiently automated and reproducible COMSOL workflow could be built for this type of parametrised breast model [22]. The resulting pipeline supports systematic comparison between model variants while keeping the model definitions and evaluation outputs traceable. The final outcome of the project is therefore a reproducible COMSOL FEM model-development pipeline for EWS-oriented breast deformation studies, together with a critical assessment of which model components are suitable for quantitative interpretation and which should still be considered exploratory model-development results.

3 Materials and Methods

3.1 Model-development workflow

3.1.1 Starting point and relation to previous EWS FEM work

This project was developed in the context of previous internal EWS finite element modelling work. Earlier EWS models provided a useful methodological starting point for breast geometry parameterisation, soft-tissue material assumptions, dynamic loading and displacement-based evaluation [23, 24]. These previous projects were used as engineering reference material and as a comparison point for modelling choices. They were not used as validation datasets for the current COMSOL model.

The present project develops a separate COMSOL-based modelling route. The motivation for this route was to improve control over constructive geometry, region definitions, model variants and post-processing while keeping the workflow reproducible. This was especially relevant because the project aimed to investigate the influence of individual anatomical and mechanical components, including chestwall shape, fibroglandular tissue distribution, skin representation, Cooper-ligament support approximation and tumor inclusion. Additional context on the previous EWS FEM work is provided in Appendix A, while the current repository structure is summarised in Appendix B.

3.1.2 COMSOL-based workflow

The finite element model was implemented in COMSOL Multiphysics and controlled using a Python-based source-case workflow [22]. Each model variant was defined by a case configuration file containing the main geometry, material, loading and evaluation settings. These settings were used to generate a consistent COMSOL model definition, including the required geometry operations, material assignments, selections, study settings and post-processing outputs. An overview of the workflow is shown in Figure 1.

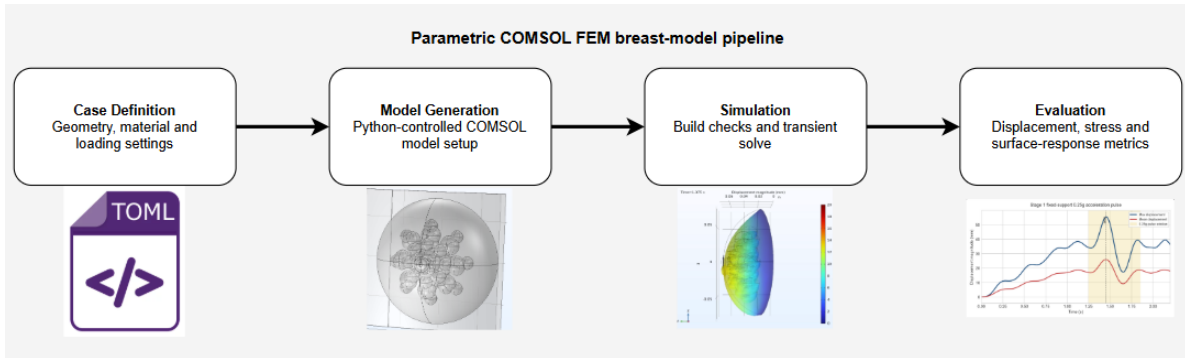


Figure 1: Overview of the COMSOL-based model-development workflow. Case definitions specify the main geometry, material and loading settings. These settings are used by the Python-controlled workflow to generate the COMSOL model, perform build and solve steps, and export compact evaluation outputs for comparison.

The purpose of this workflow was to make model changes traceable. Instead of manually modifying a COMSOL model for each comparison, the relevant modelling assumptions were stored in version-controlled case definitions. This allowed related model variants to be generated from a consistent structure and made it easier to identify which modelling change was responsible for a difference in the simulated displacement or stress response. The project repository contains the implementation of the Python workflow, the active case definitions and the compact evaluation outputs used for the comparisons [25]. A simplified example of a case definition is provided in Appendix C.

3.1.3 Model-development strategy

The model was developed by adding and testing one modelling component at a time. The development was organised around six main modelling themes: dynamic excitation, posterior chestwall geometry, internal fibroglandular tissue distribution, outer-shape variation, mechanical support representation and tumor

sensitivity. This structure was used to reduce the risk of interpreting multiple model changes at once.

The first part of the workflow focused on establishing a stable dynamic input and a controlled anatomical reference geometry. The posterior support was then refined using a curved chestwall representation, followed by a more detailed fibroglandular tissue distribution. After this anatomical route was established, additional model variants were used to investigate outer-shape asymmetry, Cooper-ligament support approximations, and skin and material sensitivity. Tumor sensitivity was treated as a separate final extension of the workflow. In this extension, spherical tumor inclusions were added to matched reference models to evaluate the effect of local stiffness changes on the simulated response. The overall model-development route is summarised in Figure 2.

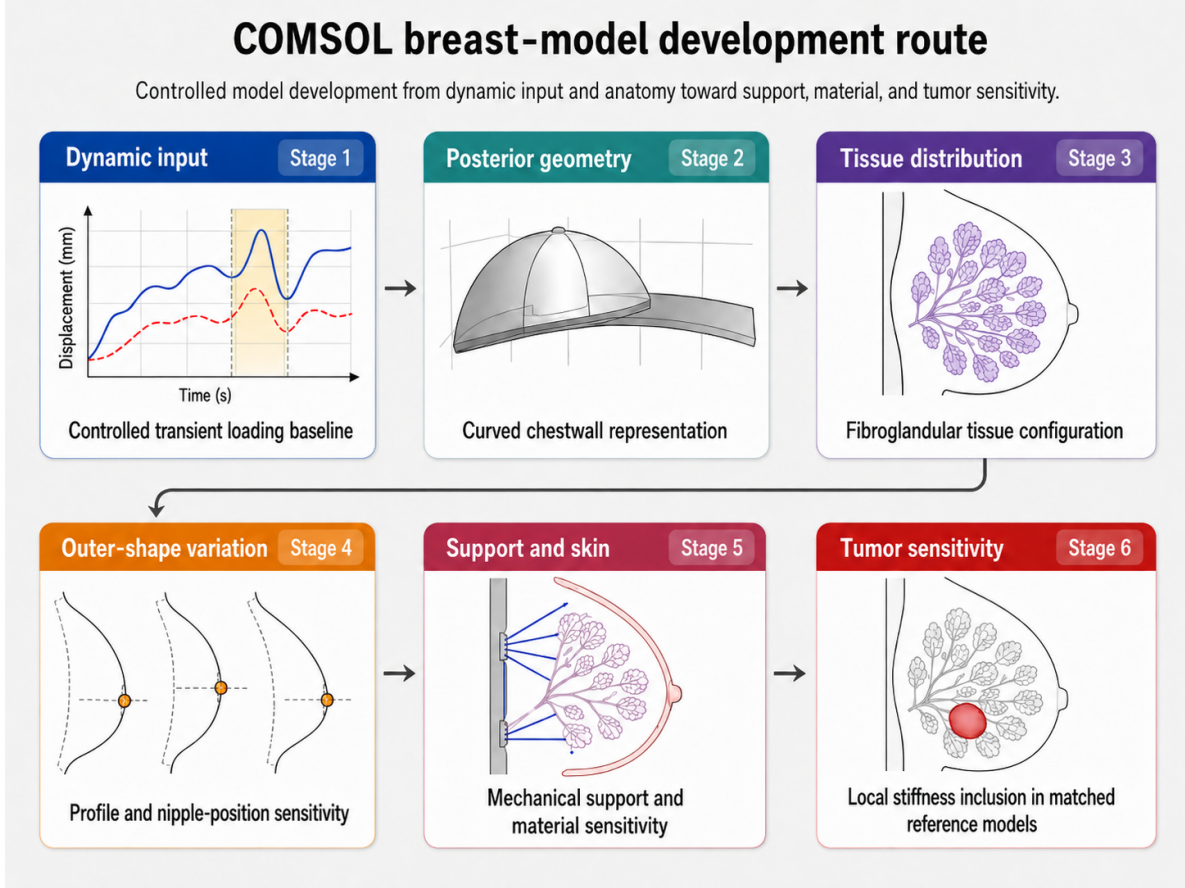


Figure 2: Schematic overview of the COMSOL breast-model development route used in this project. The workflow progresses from dynamic input and anatomical model definition toward outer-shape variation, support and skin sensitivity, and tumor sensitivity, allowing individual modelling components to be investigated in a controlled manner.

Not all model variants had the same interpretation status. Some cases were used as controlled reference models for quantitative comparison, whereas others were used as exploratory checks for numerical stability, geometry construction or sensitivity to modelling assumptions. This distinction is important because the aim of the project was not to validate a final patient-specific diagnostic model, but to construct a reproducible COMSOL FEM workflow and to identify which modelling components have a measurable influence on the simulated breast deformation response.

3.2 Anatomical model definition

The anatomical model was defined as a parametrised breast representation rather than as a reconstruction of one individual patient. This choice reflects the aim of the EWS modelling workflow to generate controlled breast-model variants that can be used to study how anatomy, tissue distribution, material assumptions and tumor properties affect the simulated surface deformation response. In the longer term, such a parametrised modelling route could support the generation of synthetic deformation data for algorithm development and AI training, provided that the simulated responses are sufficiently well understood and validated. Patient-specific breast geometries remain important for clinical modelling applications [16, 17, 19], but the present work focuses on building a reproducible model-development platform that can later be extended toward broader anatomical variation and more patient-specific inputs.

In this section, the anatomical definition is limited to the geometric and tissue-layout components of the model. This includes the parametrised breast envelope, posterior chestwall representation, fibroglandular tissue distribution, and controlled outer-shape and nipple-position variations. Anatomically motivated components that were implemented mainly through material or support definitions, such as the skin layer, Cooper-ligament support approximation and tumor inclusion, are described in Section 3.3.

3.2.1 Parametrised breast geometry and reference volume

The overall breast shape was generated from the TOML case-definition parameters used by the COMSOL pipeline. For the selected anatomical route, the model used a characteristic radius of 70 mm, an elliptic outer profile, an anterior projection scale of 1.08 and a superior-pole scale of 0.98. The radius sets the global model scale, while the profile and scaling factors adjust the anterior projection and upper-breast shape. These parameters were selected as representative model-construction settings for the reference geometry.

The reference volume was chosen by comparing the generated COMSOL geometry with literature values for breast-volume measurement and with an internal Catharina Hospital dataset of 100 manually delineated breast volumes [26–29]. The details of this comparison are provided in Appendix D. Because breast-volume estimates depend on measurement method, patient position and boundary definition, the selected volume was not treated as a single population average, but as a defensible reference size for controlled model development.

After the COMSOL geometry operations, including posterior support construction and volume-preserving compensation, the selected anatomical route produced a breast volume of ~585 mL. This value is within the lower-mid range of the internal dataset and is consistent with the literature-based volume context. It was therefore used as a controlled reference volume for the main matched comparisons. Keeping this volume fixed prevented later changes in tissue distribution, asymmetry, support representation or tumor inclusion from being combined with a change in overall breast size.

3.2.2 Posterior chestwall representation

The posterior boundary of the breast was represented by a simplified chestwall support geometry. This support was not intended as a full anatomical reconstruction of the thoracic wall or pectoral region, but as a controlled posterior interface that better reflects the curved support on which the breast rests than the flat chestwall used in earlier internal EWS modelling work. Curved posterior support is also consistent with breast FEM literature, where the breast is commonly modelled relative to the thoracic wall, pectoral region or posterior support boundary [17, 19].

In the selected reference route, the posterior support was implemented as a transverse curved support geometry with volume-preserving compensation, as shown in Figure 3. In the COMSOL case definition, this route used the support mode `transverse_vp`, a transverse curve depth of 21 mm and a chestwall-curve centre offset of +55 mm in the COMSOL x -direction. The posterior support was generated using a `cylinder_band` geometry, such that the chestwall shape could be adjusted through the curvature depth and offset parameters. A positive offset represents the selected left-breast orientation used in this report route, while the same construction can be mirrored using a negative offset for a right-breast orientation.

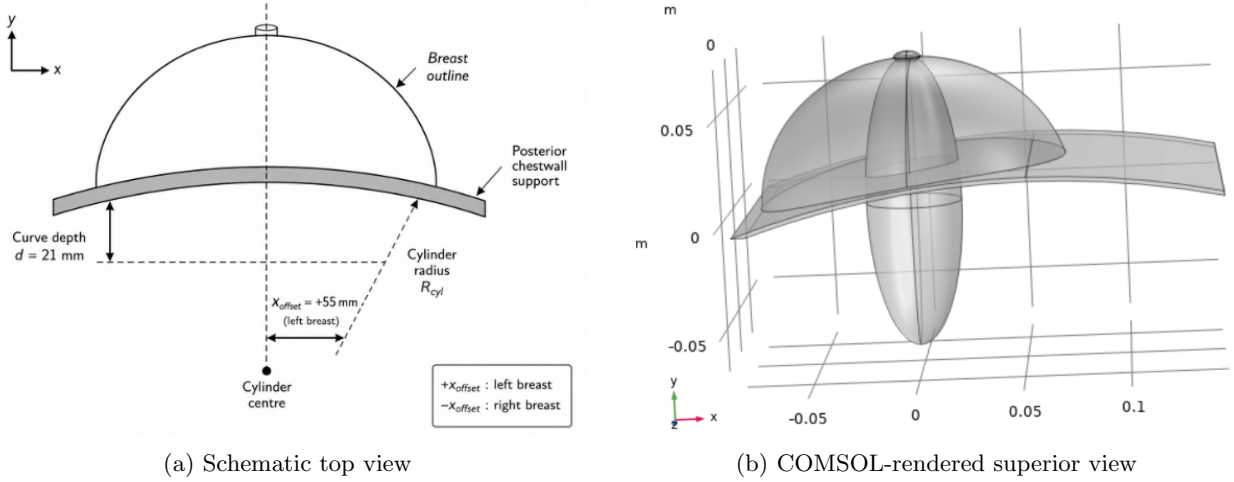


Figure 3: Posterior chestwall representation used in the selected reference route. The schematic view illustrates the transverse curved support geometry, including the curvature depth and the x -offset of the cylinder centre. The COMSOL-rendered view shows the corresponding curved posterior support geometry for the selected `xoffset055` case.

Superior–inferior curvature can also be combined with the transverse support geometry in the pipeline. However, this additional curvature was not retained in the selected reference baseline because the transverse curvature represented the dominant anatomical refinement compared with the earlier flat chestwall, while exploratory COMSOL checks showed only a limited additional effect from the superior–inferior component. Excluding this extra curvature also kept the reference geometry simpler and more stable for the matched comparisons. To preserve geometric consistency across parameter choices, the nipple-related helper geometry and anterior glandular alignment were updated automatically relative to the selected chestwall shape using projected-normal alignment.

3.2.3 Fibroglandular tissue distribution

The fibroglandular tissue representation was refined from a simple internal region toward a more structured lobule-based distribution. This step was included to make the internal tissue layout more anatomically realistic, because breast tissue is heterogeneous and internal tissue composition can influence the simulated deformation response [18, 19].

The selected reference route was guided mainly by the multi-component breast-model approach described by Chen et al. [18]. In the COMSOL implementation, the fibroglandular region was generated using the `chen_2024_template_lobes` mode. The reference configuration used 12 lobules, divided into 4 inner and 8 outer lobules, with duct-like components directed toward the nipple, a subareolar core and a subareolar bridge. A lateral view of the reference configuration is shown in Figure 4a.

Three spread variants were retained: compact, reference and wide. These variants changed the spatial spread of the lobules while keeping the selected Stage 2 chestwall geometry and the same general model setup. The reference spread was used as the main anatomical route for later comparisons, while the compact and wide variants were used to check the geometric effect of the assumed fibroglandular spread. The compact and wide variants are shown in Figure 4b.

The current reference model had a total breast volume of ~ 585 mL and used a glandular fraction of $\sim 24.3\%$, corresponding to a glandular volume of ~ 142.2 mL. This value was considered acceptable for the reference route after comparison with literature on volumetric breast density and fibroglandular tissue quantification [30–32]. Because volumetric fibroglandular fraction is not directly equivalent to projected mammographic density or to the Breast Imaging Reporting and Data System (BI-RADS) breast-composition categories, the literature context is summarised separately in Appendix E [33].

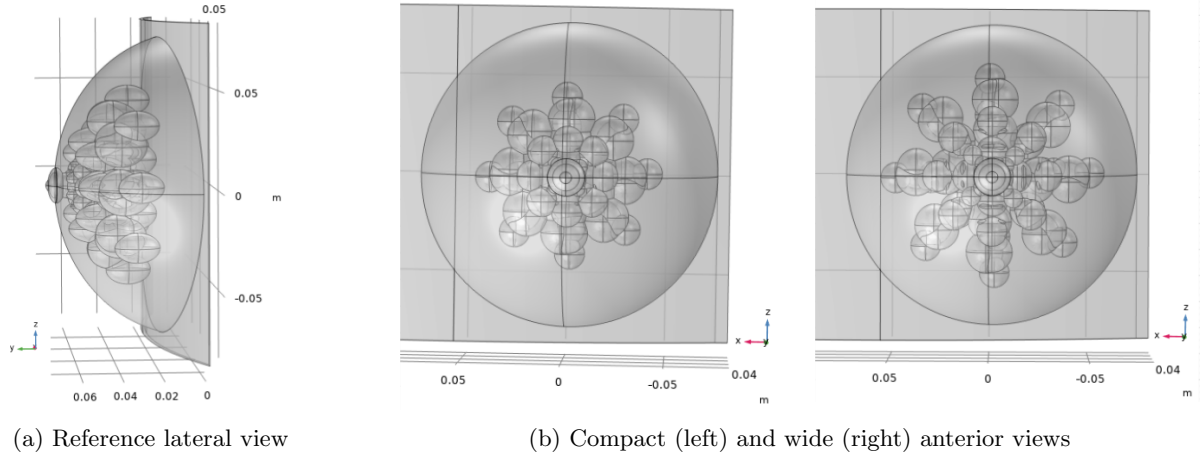


Figure 4: COMSOL-rendered fibroglandular tissue configurations used in the Stage 3 anatomical route. The reference lateral view shows the chestwall-aware lobule and duct structure inside the selected xoffset055 geometry. The compact and wide anterior views illustrate the spread variants used to evaluate the effect of fibroglandular spatial distribution.

3.2.4 Outer-shape and nipple-position variations

Outer-shape and nipple-position variations were included because breast size, shape and left–right symmetry vary substantially between individuals [19,26]. For an EWS-oriented model, this is relevant because surface shape and nipple position influence both the visible breast surface and the interpretation of local displacement or landmark motion.

Stage 4 was therefore used to test controlled geometry variations on top of the selected Stage 2 chestwall geometry and Stage 3 fibroglandular reference. In the COMSOL case definitions, the outer breast shape can be scaled in width and height, while the nipple position can be shifted over the breast surface in the x - and z -directions. Examples of these geometry variations are shown in Figure 5.

For nipple-position variations, the associated lobules, ducts, subareolar core and bridge were updated relative to the final nipple axis, so that the internal fibroglandular layout remained coupled to the shifted nipple position. These cases were used as controlled geometry-sensitivity previews and should not be interpreted as patient-specific asymmetry reconstructions.

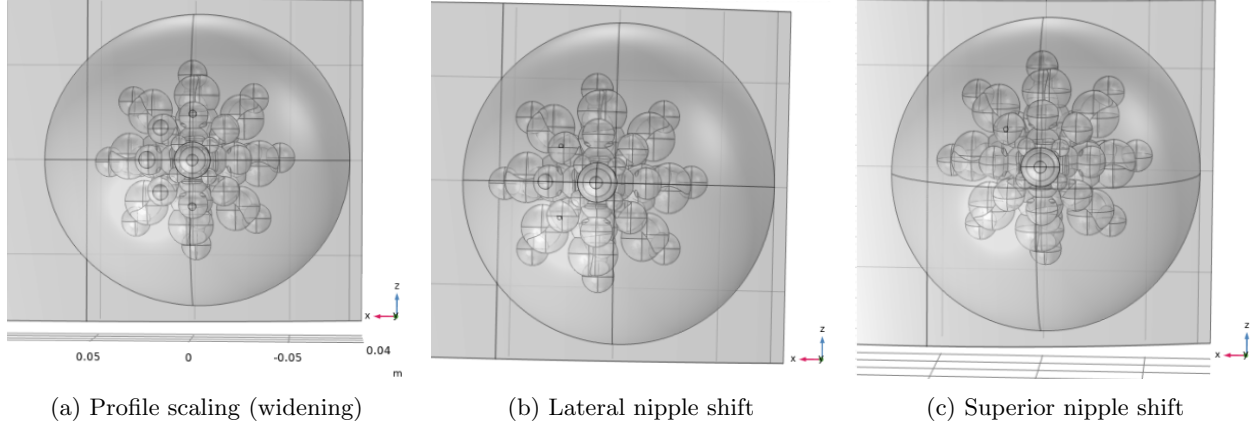


Figure 5: COMSOL-rendered examples of the outer-shape and nipple-position variation cases used in Stage 4. The cases test controlled changes in breast profile and nipple position while preserving the selected Stage 2 chestwall route and Stage 3 fibroglandular reference.

3.3 Mechanical model definition

The mechanical model defines how the anatomical regions described in Section 3.2 were assigned material behaviour and mechanical support assumptions in COMSOL. The workflow was organised into separate material and support routes, so that the influence of individual modelling choices, including tissue stiffness, skin representation, Cooper-ligament support and tumor inclusion, could be evaluated independently from the anatomical geometry. This separation allowed the mechanical contribution of each model component to be tested without redefining the underlying segmented breast shape.

3.3.1 Constitutive material model

Breast tissue was modelled as a soft, nearly incompressible material. This is consistent with common breast FEM practice, although reported mechanical properties vary substantially between studies because of differences in measurement protocol, preload, strain level, tissue type and constitutive model [4–6, 21]. In the COMSOL source-case definition, adipose, fibroglandular, skin and tumor materials were parameterised using density, bulk modulus K and two Mooney–Rivlin-type coefficients, C_{10} and C_{01} . Related Mooney–Rivlin-type soft-tissue material descriptions were also used in previous internal EWS/FEBio modelling work, which served as historical reference material for the current COMSOL route [23, 24].

The source material parameterisation can be written in terms of a deviatoric Mooney–Rivlin strain-energy density with an additional volumetric contribution [34, 35],

$$W = C_{10} (\bar{I}_1 - 3) + C_{01} (\bar{I}_2 - 3) + W_{\text{vol}}(J; K). \quad (1)$$

Here, $\bar{I}_1$ and $\bar{I}_2$ are the modified invariants, J is the volume ratio and the volumetric part is controlled by the bulk modulus K . Equation 1 describes the source material parameterisation used in the workflow. It should not be interpreted as a full derivation of the exact volumetric penalty formulation used internally by COMSOL.

For stiffness-scale interpretation, and for the effective small-strain material route used where the COMSOL implementation requires linear-elastic quantities, the Mooney–Rivlin source parameters were mapped to equivalent small-strain elastic constants using:

$$G_0 = 2(C_{10} + C_{01}), \quad (2)$$

$$E = \frac{9KG_0}{3K + G_0}, \quad (3)$$

and

$$\nu = \frac{3K - 2G_0}{2(3K + G_0)}. \quad (4)$$

The coefficients C_{10} and C_{01} determine how strongly the Mooney–Rivlin material resists shape change at small deformation. Their sum gives the small-strain shear modulus G_0 . This shear modulus was then combined with the bulk modulus K , which controls resistance to volume change, to obtain an equivalent Young’s modulus E and Poisson’s ratio ν for small-strain interpretation. Because soft breast tissue was treated as nearly incompressible, K was chosen much larger than G_0 , resulting in ν close to 0.5.

For nearly incompressible behaviour, where K is much larger than G_0 , this reduces approximately to

$$E \approx 3G_0 = 6(C_{10} + C_{01}). \quad (5)$$

The resulting stiffness value E was therefore reported as a small-strain equivalent stiffness, allowing the material sets to be compared more directly with each other and with literature values commonly reported in terms of Young’s modulus. It does not replace the original Mooney–Rivlin source-parameter definition. The posterior chestwall support was represented separately using a linear elastic surrogate material with $E = 10$ kPa and $\nu = 0.49$.

3.3.2 Material parameter sets

The material parameter sets were defined to support both a main reference configuration and additional stiffness-sensitivity cases. The reference values were selected from the literature comparison and internal model-justification notes, with the aim of keeping the adipose and fibroglandular stiffnesses within the low-kPa range commonly used for breast soft-tissue FEM models [4–6, 21]. A summary of the literature values considered during parameter selection is provided in Appendix F.

In addition to the reference set, lower- and higher-stiffness variants were tested using the same anatomical geometry and boundary-condition route. These variants were included to evaluate the sensitivity of the simulated mechanical response to the selected material stiffness.

The main material definitions used in the COMSOL model-development workflow are summarised in Table 1. The listed Young’s modulus values are included as small-strain equivalent stiffness values, so that the material scale can be compared more easily across parameter sets and with literature values reported in terms of Young’s modulus.

Table 1: Main material parameters used in the COMSOL model-development workflow.

Component	Density	Bulk / elastic setting	Material coefficients	Stiffness scale
Adipose tissue	950 kg m ⁻³	$K = 425$ kPa	$C_{10} = 310$ Pa; $C_{01} = 300$ Pa	$E \approx 3.66$ kPa
Fibroglandular tissue	1070 kg m ⁻³	$K = 425$ kPa	$C_{10} = 833$ Pa; $C_{01} = 833$ Pa	$E \approx 10.0$ kPa
Skin layer	1100 kg m ⁻³	$K = 8.33$ MPa	$C_{10} = 41667$ Pa; $C_{01} = 41667$ Pa	$E \approx 500$ kPa
Hard tumor overlay case	1079 kg m ⁻³	$K = 425$ kPa	$C_{10} = 8333$ Pa; $C_{01} = 8333$ Pa	$E \approx 100$ kPa
Chestwall support	1050 kg m ⁻³	$E = 10$ kPa; $\nu = 0.49$	Linear elastic support material	$E = 10$ kPa

3.3.3 Volumetric skin representation

The skin was treated as a separate mechanical component because its stiffness can be substantially higher than that of the internal adipose and fibroglandular tissues, making it relevant for whole-breast deformation

[21,36]. During model development, both surface-based shell and volumetric solid-layer representations were considered. Shell-based skin descriptions were used in previous internal EWS/FEBio work [23,24] and are also used in breast finite-element models reported in the literature [36]. A volumetric layer was retained for the COMSOL route because it allowed the skin to be defined through the same material parameter workflow as the internal breast tissues.

In the active skin-enabled cases, the volumetric skin layer was assigned a thickness of 1.5 mm. This value is consistent with reported breast skin thickness measurements and modelling assumptions in the literature, where average values around 1.45–1.55 mm and modelled skin thicknesses in the same range have been reported [17,36–38]. The skin material used the density, bulk modulus and Mooney–Rivlin-type coefficients listed in Table 1.

Because reported skin stiffness values differ substantially between studies and depend on the measurement method and modelling assumptions, the skin stiffness was also varied in sensitivity cases [5,21]. These variants were compared with the no-skin model to evaluate the mechanical influence of adding a distinct skin layer. Earlier COMSOL shell/scaffold skin attempts were not retained as the main implementation route because they produced unrealistic breast-surface behaviour in the generated models and were less straightforward to control than the volumetric solid-layer implementation.

3.3.4 Cooper-ligament support approximation

Cooper-ligament support was represented as a simplified mechanical support contribution rather than as a reconstructed anatomical ligament network. Similar reduced support representations are common in breast finite-element modelling, where fibrous structures such as Cooper’s ligaments are often approximated because they are difficult to segment and assign patient-specific material properties [19,21,36]. In the current COMSOL route, the support contribution was implemented through displacement-proportional restoring tractions on selected anterior support regions. This provided a separate model-development route for testing the effect of Cooper-like support on the simulated breast response.

Three support-selection variants were implemented. The first connected the nipple-region support to the posterior chestwall direction, the second applied support between skin and glandular regions, and the third used a denser skin-to-glandular support selection. These variants were used to compare different simplified approximations of anterior fibrous support without reconstructing a full anatomical Cooper-ligament network.

The support scale was defined from an assumed ligament stiffness, an active area fraction and a reference length,

$$k_{\text{Cooper}} = \frac{E_{\text{Cooper}} f_{\text{area}}}{L_{\text{ref}}}. \quad (6)$$

The active restoring traction was applied in the global anterior–posterior direction as

$$t_y = -k_y v, \quad (7)$$

where v is the COMSOL displacement component in the y -direction and k_y is the corresponding support scale. For the selection-specific support regions, the implemented scaling factors were

$$k_{\text{skin}} = 0.65k_{\text{Cooper}}, \quad k_{\text{gland}} = 0.45k_{\text{Cooper}}, \quad k_{\text{nipple}} = 0.20k_{\text{Cooper}}. \quad (8)$$

The default support scale used $E_{\text{Cooper}} = 5.8$ MPa, $f_{\text{area}} = 0.12$ and $L_{\text{ref}} = 0.04$ m, giving $k_{\text{Cooper}} = 17.4 \times 10^6$ N m^{−3}. Lower and higher area-fraction variants were also prepared as sensitivity cases.

Because reported Cooper-ligament properties and effective whole-breast support values differ substantially between studies, the current values should be interpreted as support-strength parameters rather than direct anatomical ligament measurements [21,36]. The no-Cooper model was retained as the comparison route,

while the Cooper-support variants were implemented to evaluate the additional mechanical influence of simplified Cooper-like support.

3.3.5 Tumor representation

Tumor inclusion was implemented as an analytic spherical material overlay inside the existing breast volume. The tumor was therefore not defined as a separate COMSOL geometry domain, but as a local modification of the material density and stiffness parameters inside the existing adipose and fibroglandular domains.

The tumor mask was defined as

$$m_{\text{tumor}}(\mathbf{x}) = \begin{cases} 1, & \|\mathbf{x} - \mathbf{x}_{\text{tumor}}\| \leq r_{\text{tumor}}, \\ 0, & \|\mathbf{x} - \mathbf{x}_{\text{tumor}}\| > r_{\text{tumor}}. \end{cases} \quad (9)$$

Here, $\mathbf{x}_{\text{tumor}}$ is the tumor centre and r_{tumor} is the tumor radius. In the COMSOL implementation, this corresponds to the scalar variable `tumor_mask`, which is equal to one inside the spherical overlay and zero outside it.

The local effective material properties were then modified using the tumor mask. For a surrounding tissue with base density ρ , base Young's modulus E and base Mooney–Rivlin coefficients C_{10} and C_{01} , the effective properties were defined as

$$\rho_{\text{eff}} = \rho + m_{\text{tumor}}(\rho_{\text{tumor}} - \rho), \quad (10)$$

$$E_{\text{eff}} = E + m_{\text{tumor}}(E_{\text{tumor}} - E), \quad (11)$$

$$C_{10,\text{eff}} = C_{10} + m_{\text{tumor}}\Delta C_{10,\text{tumor}}, \quad (12)$$

and

$$C_{01,\text{eff}} = C_{01} + m_{\text{tumor}}\Delta C_{01,\text{tumor}}. \quad (13)$$

The current tumor material settings include both modest stiffness-contrast cases and a harder diagnostic-contrast case. Because the Early Warning Scan concept is aimed at detecting malignant breast tumors, the harder tumor-overlay setting was included to test a clear local stiffness contrast relative to normal breast tissue. In this high-contrast setting, the analytic overlay used $E_{\text{tumor}} = 100$ kPa together with local Mooney–Rivlin coefficient increments of $\Delta C_{10,\text{tumor}} = \Delta C_{01,\text{tumor}} = 8333$ Pa. These values should be interpreted as a simplified stiffness target for a sensitivity case rather than as a patient-specific tumor material definition, because malignant breast tumor stiffness values reported in elastography and mechanical testing can vary strongly with measurement modality, region of interest and tumor type [4, 7, 8, 39].

3.4 Dynamic loading conditions

Dynamic loading was used to excite the breast model after the mechanical geometry, material parameters and posterior support definition had been established. The loading was not intended to reproduce a complete body-motion experiment, but to provide a controlled excitation for comparing model variants. This follows the use of simplified dynamic inputs in breast-motion modelling, where finite-element models are used to study how tissue composition, support and activity-related excitation influence breast deformation [21, 36].

3.4.1 Fixed-support acceleration input

The fixed-support acceleration route used a stationary posterior support while applying an additional vertical body acceleration to the breast volume. This represents breast motion relative to a fixed torso/support frame and was used to test how the model responded to controlled acceleration amplitudes.

Before the dynamic phase, gravity was ramped to establish a preloaded gravitational deformation state. After a short hold period, the additional acceleration pulse was applied using a smooth full-sine profile,

$$a_{\text{pulse},z}(t) = \begin{cases} -A_g g \sin\left(2\pi \frac{t-t_0}{T}\right), & t_0 \leq t \leq t_0 + T, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

Here, A_g is the acceleration amplitude expressed as a fraction of gravitational acceleration, $g = 9.81 \text{ ms}^{-2}$, T is the pulse duration and t_0 is the start time of the dynamic pulse. The term $A_g g$ therefore gives the peak additional acceleration in ms^{-2} . The mild acceleration case used $A_g = 0.25$ and $T = 0.60 \text{ s}$, corresponding to a peak additional acceleration of approximately 2.45 ms^{-2} . Higher-amplitude cases were also tested to evaluate the sensitivity of the simulated response to the selected acceleration input.

A Rayleigh damping formulation was used with a mass-proportional coefficient $\alpha = 60 \text{ s}^{-1}$ and no stiffness-proportional damping term,

$$\mathbf{C} = \alpha \mathbf{M} + \beta \mathbf{K}, \quad \beta = 0. \quad (15)$$

This damping was included to reduce free oscillations after the acceleration pulse. The acceleration amplitude and damping settings were treated as modelling parameters for controlled dynamic comparison, not as patient-specific motion measurements.

3.4.2 Gravity preload and prescribed support displacement

The gravity-preload step was used in both dynamic routes to initialise the model from a loaded mechanical state. This is relevant because the breast response depends on the gravitational equilibrium configuration, especially for soft-tissue models with low stiffness and nearly incompressible material behaviour [19, 21].

In addition to the fixed-support acceleration route, a prescribed support-displacement route was implemented as a separate motion variant. In this route, the posterior attachment boundary was assigned a smooth vertical displacement. This is easier to relate to a person moving upward and downward, for example during a jump-like torso motion, because the support boundary itself is displaced rather than only applying an acceleration field.

The prescribed displacement was defined using a smooth bump function,

$$z_{\text{support}}(t) = \begin{cases} A 64 s^3 (1-s)^3, & 0 \leq s \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad s = \frac{t-t_0}{T}. \quad (16)$$

Here, A is the prescribed displacement amplitude of the support and T is the duration of the motion. The factor 64 normalises the bump function so that its maximum equals A , because $s^3(1-s)^3$ reaches $1/64$ at $s = 0.5$. The function also starts and ends smoothly, avoiding abrupt displacement jump at the beginning or end of the prescribed motion.

Various displacement-support variants with amplitudes of 20 mm, 40 mm and 60 mm over 0.60 s were prepared to test how explicit posterior support motion affected breast-surface response. For these cases, both absolute and support-relative displacement measures were considered. The relative vertical displacement of the support was defined as

$$w_{\text{rel}}(t) = w(t) - z_{\text{support}}(t), \quad (17)$$

where $w(t)$ is the vertical displacement of the evaluated breast surface. This measure separates deformation of the breast relative to the moving support from the imposed support motion itself.

3.5 Mesh, build and solver strategy

The COMSOL models were generated from scripted case definitions. Each simulation case was stored as a TOML file and translated into a COMSOL Java builder script, so that geometry settings, material parameters, loading conditions and solver settings remained reproducible between model variants.

Meshing was performed in COMSOL using a physics-controlled free tetrahedral mesh. The main reported cases used a characteristic mesh length of 5 mm with second-order solid-mechanics discretisation. The same mesh settings were used across comparable cases to avoid introducing mesh-dependent differences between model variants. The mesh was therefore treated as a fixed modelling choice for matched model comparisons; a full mesh-convergence study was outside the scope of the present model-development work. An example of the generated mesh is shown in Figure 6.

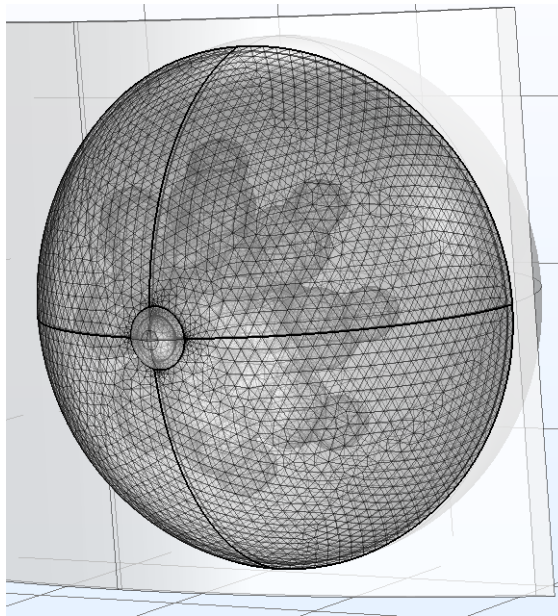


Figure 6: Example of the generated COMSOL tetrahedral mesh used for the breast model.

The solver strategy consisted of a gravity-preload step followed by the time-dependent dynamic simulation. Build-only cases were used during development to verify generated selections and model setup before running full dynamic simulations.

3.6 Post-processing and evaluation metrics

Post-processing was performed using the scripted COMSOL export and Python analysis workflow. For each completed simulation, compact output files were generated for displacement fields, selected stress quantities and time-dependent response measures. These outputs were used to evaluate how changes in model settings or parameter choices affected the simulated mechanical response, using the corresponding comparison case for each analysis.

Because the Early Warning Scan concept is based on observable breast-surface motion, the main evaluation focused on the free outer breast surface. In the COMSOL workflow this surface was represented by the `outer_skin_free_bnd` selection. Whole volume displacement and stress measures were retained as supporting diagnostic quantities.

The displacement magnitude was defined as

$$u_{\text{mag}} = \sqrt{u^2 + v^2 + w^2}, \quad (18)$$

where u , v and w are the displacement components in the x -, y - and z -directions. In addition to the magnitude, signed displacement components were evaluated because they preserve direction. The vertical displacement component w was used to describe upward or downward surface motion during dynamic excitation.

For surface-based evaluation, spatial averages were computed over the evaluated surface selection S ,

$$\bar{q}_S(t) = \frac{1}{A_S} \int_S q(\mathbf{x}, t) dA, \quad (19)$$

where q is the evaluated displacement or stress quantity and A_S is the surface area of the selection. For dynamic cases, signed vertical displacement was also evaluated relative to the dynamic-start state,

$$\Delta \bar{w}_S(t) = \bar{w}_S(t) - \bar{w}_S(t_{\text{dyn, start}}). \quad (20)$$

For prescribed support-displacement cases, the support-relative vertical displacement defined in Equation 17 was used in addition to the absolute displacement. Landmark-patch displacements were also exported for selected surface regions, including the nipple and superior, inferior, left and right breast-surface patches. These patch-averaged values were used as robust surface-motion measures, because they are less sensitive to individual mesh nodes than single-point displacement values.

Stress was evaluated using the exported von Mises stress field. This was used as a mechanical diagnostic for comparing stress concentration between model variants. For the tumor cases, local displacement and stress measures were additionally evaluated using the tumor mask region.

Table 2: Post-processing quantities used to compare COMSOL model variants.

Quantity	Definition or source	Use in evaluation
Free-surface displacement	u , v , w and u_{mag} on <code>outer_skin_free.bnd</code>	Main EWS-relevant surface-motion response.
Surface-average response	$\bar{q}_S(t)$ and $\Delta \bar{w}_S(t)$	Average dynamic response of the visible breast surface.
Landmark-patch displacement	Patch-averaged displacement on nipple and breast-surface landmark selections	Robust comparison of visible local surface motion.
Support-relative displacement	$w_{\text{rel}} = w - z_{\text{support}}$	Deformation relative to prescribed support motion.
von Mises stress	Exported COMSOL stress field	Mechanical stress-concentration diagnostic.
Tumor-region response	Mask-based displacement and stress measures	Local response around the analytic tumor overlay.
Peak response	Maximum value over the evaluated time range	Quantitative comparison between corresponding cases.

When variants were compared, changes were reported relative to the corresponding comparison case,

$$\Delta q_{\%} = 100 \frac{q_{\text{variant}} - q_{\text{comparison}}}{q_{\text{comparison}}}, \quad (21)$$

where q denotes the evaluated displacement or stress metric. This percentage difference was used only when the comparison value was non-zero and physically meaningful.

3.7 Reproducibility and data management

The project was organised to keep the COMSOL workflow reproducible and traceable. Source code, TOML case definitions, report notes and compact post-processing outputs were maintained in the project repository [25]. Large generated files, such as full COMSOL model files, temporary solver files and extensive raw exports, were kept outside version control because they can be regenerated from the scripted case definitions.

Each COMSOL case was defined by a version-controlled TOML file. These files stored the relevant geometry route, material parameters, loading settings, mesh settings and output options for each model variant. The Python pipeline converted these case definitions into COMSOL Java builder scripts and post-processing outputs. This reduced manual editing in COMSOL and made it easier to inspect or rerun model variants.

The simulations were run using a licensed COMSOL Multiphysics installation available through TU/e. All model builds and time-dependent solves reported in this work were executed on local hardware during model development. The software and hardware environment used for the simulations is summarised in Table 3.

Table 3: Software and hardware environment used for COMSOL model development and simulation runs.

Item	Setting
COMSOL version	COMSOL Multiphysics 6.4
License environment	TU/e licensed COMSOL installation
Operating system	Windows 11 Enterprise
Processor	11th Gen Intel Core i7-11800H, 2.30GHz
Memory	16.0 GB RAM
Python environment	Python 3.8.8
Repository	[25]

4 Results

4.1 Overview of completed model variants

The Results chapter focuses on the model variants that were generated to evaluate the developed COMSOL breast model. Table 4 summarises which result groups were available and how they are used in the following subsections. This distinction is included because some routes produced complete quantitative outputs, while others were retained as scout, diagnostic or build-level outputs.

Table 4: Overview of model-variant groups used in the Results chapter.

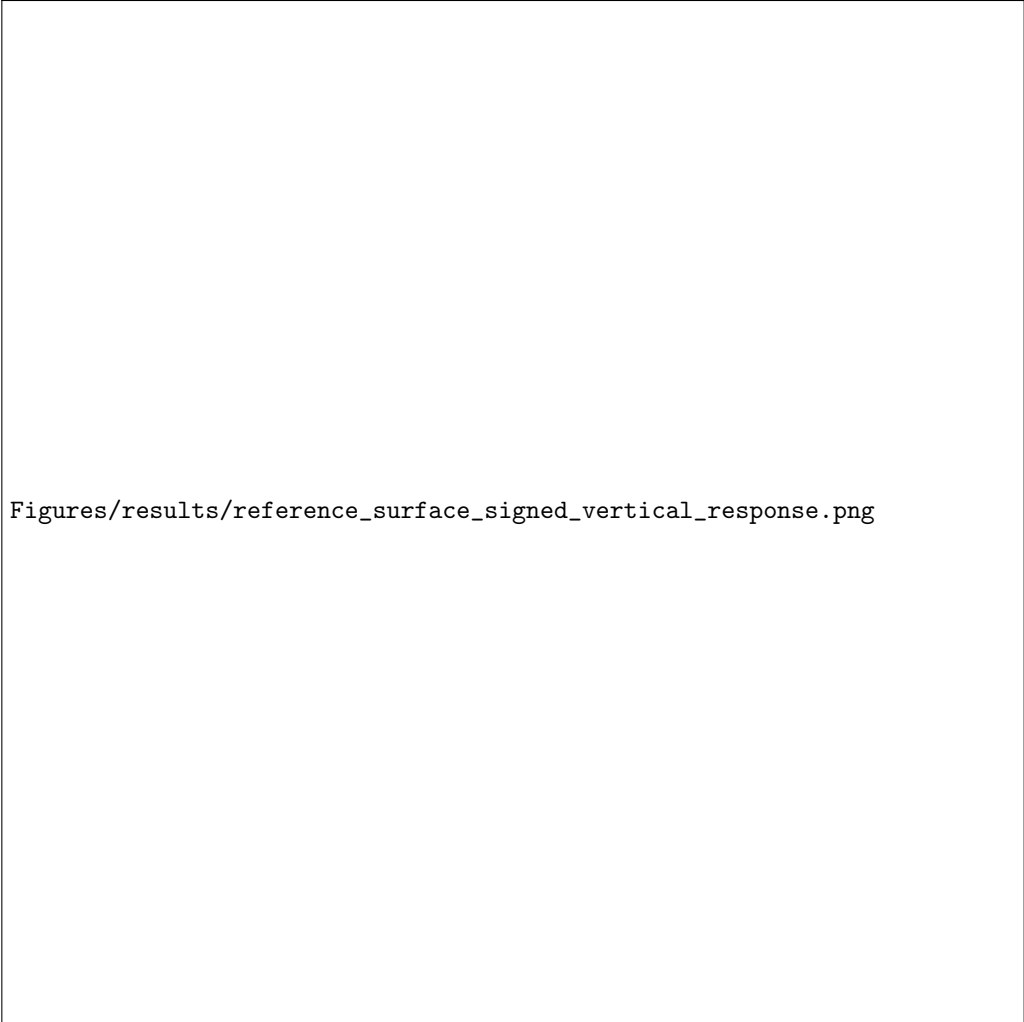
Result group	Available output	Use in this Results chapter
Reference anatomical model	Solved anatomical route with selected chestwall geometry and realistic glandular distribution	Establishes the main anatomical model used for later comparisons
Material and skin variants	Quantitative displacement and stress outputs for selected material and volumetric-skin cases	Used to compare material and skin-related response changes
Cooper-like support	No-Cooper comparison case and Cooper-like support scout outputs	Used to report current support-sensitivity status
Dynamic loading	Solved fixed-support acceleration cases and prescribed support-displacement output	Used to compare loading-route response measures
Tumor overlay	Build-level tumor placement examples	Used to show successful tumor-mask implementation and placement

The following subsections report quantitative trends only when the output status supports such comparison. Diagnostic, build-level or incomplete cases are reported as implementation status rather than as final quantitative trends.

4.2 Reference model response

The developed COMSOL workflow produced a solvable anatomical breast model with the selected curved chestwall geometry and realistic glandular distribution. This model was used as the anatomical basis for the sensitivity comparisons in the following subsections. The reference route retained a breast volume close to 585 mL and increased the glandular fraction to approximately 24.3% after introducing the realistic glandular tissue distribution.

The generated analysis outputs included displacement fields, signed vertical surface response, displacement statistics and von Mises stress fields. An example of the exported surface-response output is shown in Figure 7. These outputs were used to check that the model produced a measurable dynamic response under the fixed-support acceleration input before individual material, skin, Cooper-like support, loading and tumor-overlay variants were compared.



Figures/results/reference_surface_signed_vertical_response.png

Figure 7: Example of the signed vertical surface-response output generated for the anatomical reference route. The figure illustrates the type of surface-motion metric used for later comparison between model variants.

Time-dependent displacement and stress visualisations were also generated during post-processing. These outputs were used for visual inspection of the deformation pattern and local stress distribution, while the quantitative comparisons in the following subsections are based on the exported displacement and stress metrics.

4.3 Effect of material-parameter variation

Material-parameter variation was evaluated using skin-enabled full-glandular model variants with different internal tissue and skin stiffness settings. The peak displacement and von Mises stress values are summarised in Table 5.

Table 5: Material-parameter variation results for skin-enabled model variants. VM denotes von Mises stress.

Case	Peak avg. disp. (mm)	Peak max. disp. (mm)	Peak avg. VM (kPa)	Peak max. VM (kPa)
Volumetric skin, 1.25g	3.40	5.93	1.38	51.63
Volumetric skin + soft interior, 1.25g	5.90	10.90	1.80	72.38
Volumetric skin 88 kPa + soft interior, 1.25g	13.70	24.84	1.30	44.31
0.1 mm soft skin + soft interior, 1.25g	27.20	60.03	0.96	44.58

Within the skin-enabled variants, the peak maximum displacement increased from 5.93 mm for the volumetric-skin case to 10.90 mm when the internal tissue stiffness was reduced. The lower-stiffness skin variants produced larger peak maximum displacements of 24.84 mm and 60.03 mm. Peak maximum von Mises stress ranged from 44.31 kPa to 72.38 kPa across the same material-parameter set.

4.4 Effect of volumetric skin representation

The volumetric skin representation was evaluated on the same full-glandular model route by comparing the no-skin case with skin-enabled variants at the same 1.25g dynamic input. The skin-enabled cases used a 1.5 mm volumetric skin layer. The peak displacement and von Mises stress values are summarised in Table 6.

Table 6: Comparison of no-skin and 1.5 mm volumetric-skin model variants at 1.25g. VM denotes von Mises stress.

Case	Peak avg. disp. (mm)	Peak max. disp. (mm)	Peak avg. VM (kPa)	Peak max. VM (kPa)
No skin, 1.25g	7.34	14.23	0.80	3.92
Volumetric skin, 1.25g	3.40	5.93	1.38	51.63
Volumetric skin + soft interior, 1.25g	5.90	10.90	1.80	72.38

Compared with the no-skin case, the volumetric-skin case reduced peak average displacement from 7.34 mm to 3.40 mm and peak maximum displacement from 14.23 mm to 5.93 mm. With the soft-interior material variant, peak average displacement increased to 5.90 mm and peak maximum displacement increased to 10.90 mm. Peak maximum von Mises stress was higher in both skin-enabled variants than in the no-skin case.

An example von Mises stress field for the Stage 5 no-Cooper 1.25g case is shown in Figure 8. The highest stress values in this example were located near the posterior chestwall/support attachment region.

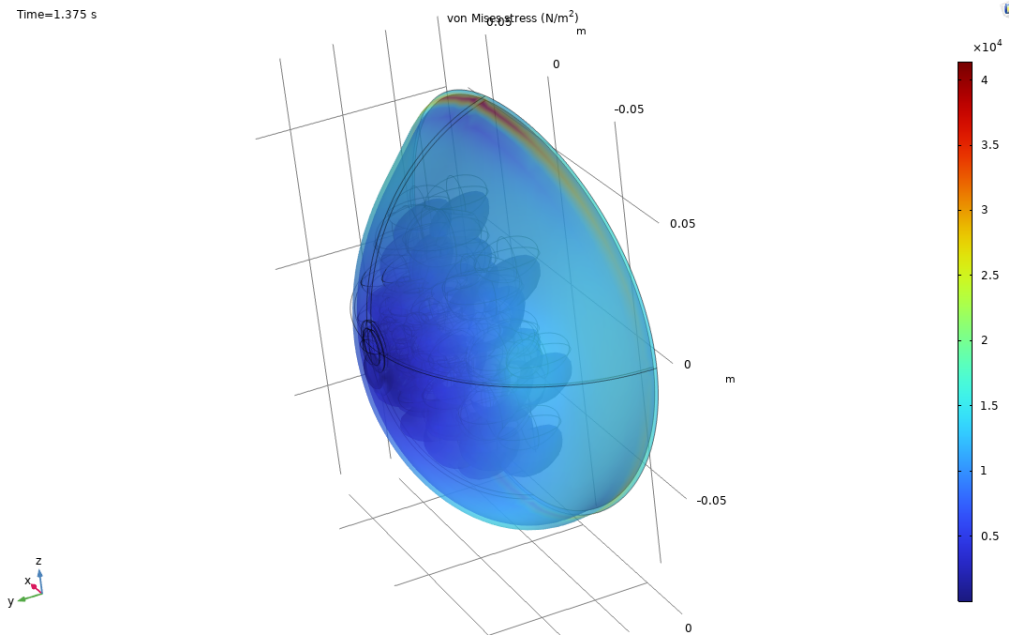


Figure 8: Example von Mises stress distribution for the Stage 5 no-Cooper 1.25g case at $t = 1.375$ s. The highest stress values are located near the posterior chestwall/support attachment region, while the free breast volume shows lower stress values.

4.5 Effect of Cooper-like support

The Cooper-like support approximation was first evaluated against the no-Cooper comparison case. In the anatomical Stage 5 route, the no-Cooper case reached the expected final time of 2.2 s, while the default Cooper-support case failed immediately during the dynamic solve because of solver-capacity limitations. The failed case was therefore not used for quantitative response comparison.

A separate simple-gland Cooper-support scout was used to compare the implemented support-selection variants. The resulting peak displacement values are summarised in Table 7.

Table 7: Cooper-like support scout results for the simple-gland model route.

Case	Peak avg. disp. (mm)	Peak max. disp. (mm)	Avg. disp. change (%)	Max. disp. change (%)
No Cooper baseline	22.55	45.58	0.00	0.00
Gland-to-skin area 0.03	22.44	44.99	-0.50	-1.29
Gland-to-skin area 0.06	22.44	44.97	-0.52	-1.34
Gland-to-skin area 0.12	22.43	44.96	-0.53	-1.37
Nipple-to-chest area 0.03	22.55	45.64	-0.03	0.12
Dense network area 0.03	22.44	44.99	-0.50	-1.29

In the simple-gland scout, the gland-to-skin support variants changed peak average displacement by -0.50% to -0.53% and peak maximum displacement by -1.29% to -1.37% relative to the no-Cooper baseline. The nipple-to-chest variant changed peak average displacement by -0.03% and peak maximum displacement by 0.12% .

4.6 Effect of dynamic loading route

4.6.1 Fixed-support acceleration amplitude

The fixed-support acceleration response was evaluated on the no-Cooper full-glandular model route. The 0.25g and 0.50g cases reached the expected final time of 2.2 s, while the 0.75g case did not produce a valid solved output. The compact review metrics are summarised in Table 8.

Table 8: Dynamic-amplitude results for the fixed-support acceleration route.

Case	End time (s)	Review avg. disp. (mm)	Surface signed w (mm)	Output status
No Cooper xoffset055, 0.25g	2.20	3.00	0.161	Solved
No Cooper xoffset055, 0.50g	2.20	2.86	0.310	Solved
No Cooper xoffset055, 0.75g	–	–	–	Failed/not valid

The surface signed vertical response increased from 0.161 mm for the 0.25g case to 0.310 mm for the 0.50g case. The no-skin 1.00g and 1.25g cases were evaluated in the separate skin/material scout, which used a simple-glandular model basis. They are therefore not included in the fixed-support amplitude comparison in Table 8.

4.6.2 Prescribed support-displacement route

A prescribed support-displacement route was evaluated as a separate motion variant. This case used the xoffset055 chestwall geometry with a simple glandular structure, no Cooper-like support, a 1.5 mm volumetric skin layer and soft internal material settings. For the 40 mm, 0.60 s support-displacement case, support-relative displacement metrics were extracted from the free outer surface and nipple landmark region.

Table 9: Prescribed support-displacement response for the 40 mm, 0.60 s motion variant.

Metric	Maximum value	Selection
Support-relative spatial average of w	8.81 mm	Free outer surface
Support-relative maximum of w	15.70 mm	Free outer surface
Support-relative spatial average of w	15.51 mm	Nipple landmark region

4.7 Tumor-overlay sensitivity

The tumor-overlay route was successfully implemented as a build-level model variant. The analytic tumor mask could be placed inside the existing breast volume and visualised for different tumor positions and sizes. Figure 9 shows representative examples of central, upper-outer surface-proximal and subareolar tumor-overlay placements.

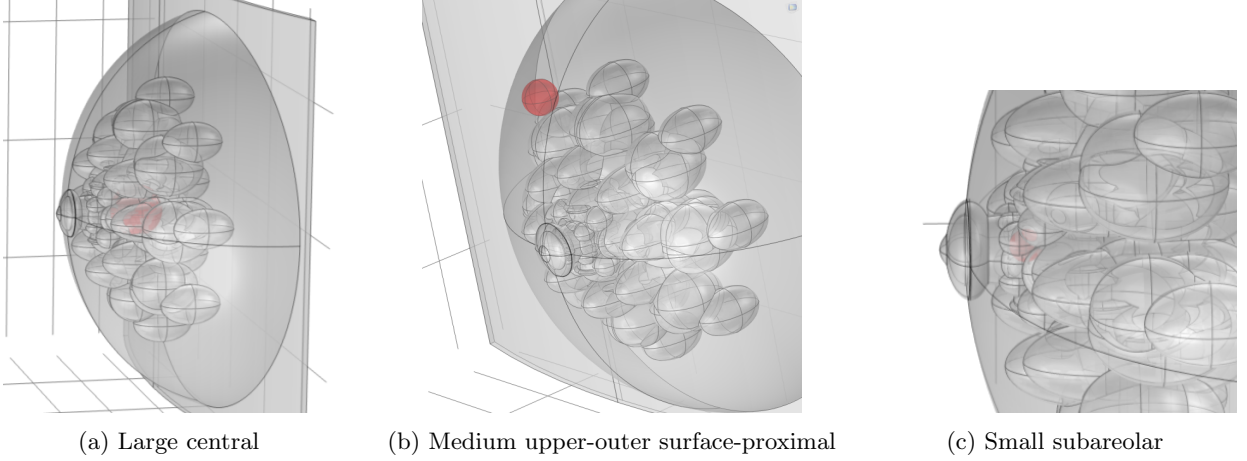


Figure 9: Representative build-level tumor-overlay placements generated in COMSOL. The breast volume and glandular structures are shown semi-transparently, while the analytic spherical tumor overlay is highlighted in red. These views show that the tumor mask could be positioned inside the existing anatomical breast model for different locations and sizes.

The build-level tumor variants included central, upper-outer, subareolar, posterior and surface-proximal placements. These variants confirmed that the tumor mask could be positioned and visually inspected within the anatomical breast model. Due to local solver-capacity and project-time limitations, complete dynamic tumor-overlay solves were not included as final quantitative result cases in this report.

The tumor-overlay results in this chapter are therefore limited to successful build-level placement examples. Complete dynamic tumor-overlay sensitivity results require a larger solved case set with complete tumor-mask post-processing.

4.8 Summary of main result trends

The Results show that the COMSOL workflow produced a more anatomically complete breast model route and several controlled model variants. The selected chestwall geometry and realistic glandular distribution provided the anatomical basis for the later sensitivity comparisons. On top of this reference route, material-parameter changes, the 1.5 mm volumetric skin layer, Cooper-like support definitions, dynamic loading variants and analytic tumor-overlay placements were implemented as separate model-development routes.

The strongest quantitative result sets were obtained for the material, skin and dynamic-loading comparisons. Lower-stiffness material settings increased the reported displacement amplitudes in the skin-enabled variants, while the 1.5 mm volumetric skin layer reduced displacement compared with the no-skin case at the same 1.25g input. For the dynamic loading route, the 0.25g and 0.50g fixed-support acceleration cases solved successfully, and the prescribed support-displacement route produced measurable support-relative surface displacement.

The Cooper-like support and tumor-overlay routes added additional model functionality, but were reported with their current evidence status. Cooper-like support was available as a simple-gland scout and as an anatomical implementation route, while the default anatomical Cooper-support solve was not retained as a final quantitative result. The tumor-overlay route was successfully generated and visualised at build level, but complete dynamic tumor-overlay sensitivity results were not included as final quantitative result cases.

5 Discussion

5.1 Contribution to an EWS-oriented FEM modelling workflow

This project developed a progressively more complete COMSOL finite-element route for simulating dynamic breast deformation in the context of the Early Warning Scan (EWS). The main contribution was not a clinically validated detection model, but a reproducible model-development framework in which anatomical structure, material parameters, support assumptions, loading route and tumor-overlay placement can be varied in a controlled way. This distinction is important because EWS is based on observable breast-surface motion, whereas many of the model quantities used during development, such as internal stress or volume-averaged displacement, are primarily mechanical diagnostics.

Compared with a simplified breast representation, the final model route introduced a curved posterior support geometry, a more realistic glandular distribution, optional volumetric skin representation, material-parameter sensitivity cases, Cooper-like support approximations, dynamic loading variants and analytic tumor-overlay placements. These additions increased the anatomical and mechanical richness of the model and made it possible to evaluate which components most strongly affect the simulated surface response. The framework therefore provides a basis for future EWS-oriented surface-motion studies, but the present results should be interpreted as model-development outcomes rather than as evidence of diagnostic performance.

5.2 Anatomical realism and glandular representation

A central aim of the project was to move toward a more anatomically realistic FEM breast model. The realistic glandular route improved on a simple glandular approximation by introducing a spatially distributed fibroglandular structure within the breast volume. This is relevant because breast tissue is mechanically heterogeneous: adipose and fibroglandular tissue differ in stiffness, density and spatial distribution, and the resulting deformation response can depend on both the amount and location of fibroglandular tissue [4–6,21].

The current glandular implementation also introduced a practical limitation. A detailed glandular structure is more representative than a single simple inclusion, but it increases geometric and meshing complexity and therefore solver cost. Future work should investigate whether a reduced glandular representation can preserve the relevant mechanical effects while being faster to mesh and solve. Possible routes include controlled glandular-fraction models, smoothed glandular masks, clustered glandular regions or subject-specific segmentations with simplified surfaces. Such comparisons would help determine whether the full glandular structure is required for EWS-relevant surface-motion prediction, or whether a computationally cheaper approximation is sufficient for parameter sweeps.

Anatomical variability should also be included more systematically. Breast size, glandular fraction, glandular asymmetry, posterior attachment geometry and tissue distribution vary strongly between individuals. The current model should therefore be interpreted as one anatomical route rather than a population-level model. A stronger future study would run the same loading and material assumptions across multiple anatomies and would separate the influence of geometry from the influence of material parameters.

5.3 Material-parameter uncertainty and response realism

Material parameters remain one of the largest sources of uncertainty in breast FEM modelling. Reported stiffness values for adipose tissue, fibroglandular tissue, skin and tumor tissue vary substantially across studies because they depend on measurement method, tissue preparation, preload, strain level, loading direction, constitutive model and pathology [4–6,21]. The material settings used in this project should therefore be interpreted as controlled reference and sensitivity values rather than as patient-specific tissue properties.

This uncertainty affects how the displacement results should be read. A simulated displacement amplitude is only physically meaningful when the material parameters, boundary conditions and loading route are all plausible for the intended experimental situation. The current results show that the model can produce measurable dynamic surface motion, but the exact displacement scale still requires comparison with experimental or EWS-like motion data. In future work, the simulated surface displacements should be compared with phantom measurements, optical surface tracking or in-vivo breast-motion data under a known loading condition. Such validation would make it possible to determine whether the selected stiffness range reproduces realistic breast motion.

The von Mises stress results should be interpreted even more cautiously. In this project, von Mises stress was useful as a scalar diagnostic for comparing stress concentrations between variants, especially near support and attachment regions. However, stress is not directly measured by the EWS concept and should not be treated as a direct detection output. Stress concentrations near the posterior support are more likely to reflect the imposed boundary condition and support geometry than a clinically meaningful tissue response. For EWS interpretation, free-surface displacement and landmark motion are therefore more relevant than internal stress metrics.

5.4 Dynamic loading and surface-motion interpretation

The dynamic loading route is another key modelling choice. The fixed-support acceleration input and the prescribed support-displacement route represent different idealisations of how breast motion could be generated during an EWS-like measurement. The fixed-support acceleration route is useful because it applies a controlled global excitation while keeping the support geometry fixed. The prescribed support-displacement route is more directly related to a physical situation in which the chestwall or support moves up and down, for example during a jump-like motion.

Further work is needed to determine which representation best matches the intended EWS measurement setting. Relevant parameters include acceleration amplitude, pulse duration, frequency content, damping, gravity preload, support-displacement amplitude and coupling between the support and breast volume. The most appropriate dynamic input should be selected based on measurable surface response, not only numerical stability. Because EWS observes external motion, the main comparison quantity should remain the displacement of the free breast surface, supported by landmark regions such as the nipple and other visible surface patches.

The current dynamic results should therefore be interpreted as proof that different loading routes can be implemented and compared within the COMSOL workflow. They do not yet define an experimentally validated EWS loading protocol. A future validation study should record the motion applied in an EWS-like setup and use that measured motion as the input or calibration target for the FEM model.

5.5 Skin and Cooper-like support modelling choices

The skin representation had a clear mechanical role in the model because the skin layer adds a relatively stiff outer structure around softer internal breast tissue. This is consistent with multi-component breast modelling approaches in which skin, adipose tissue, fibroglandular tissue, pectoral support and fibrous support structures are treated as separate mechanical components [19, 21, 36]. The current COMSOL route used a 1.5 mm volumetric skin layer as a controlled modelling choice. This improved interpretability compared with earlier shell or scaffold attempts, but it is still not a patient-specific skin reconstruction.

The skin route should be developed further by varying both thickness and stiffness. Skin thickness and mechanical properties vary between individuals and body locations, and the effective contribution of skin in a breast model depends on how it is coupled to the internal tissue. A future parameter sweep should therefore test whether the same qualitative response is obtained for a realistic range of skin thicknesses and stiffnesses. This would also help separate the mechanical effect of a true skin layer from numerical effects caused by a stiff outer domain.

The Cooper-like support implementation should be interpreted as a simplified support approximation. Cooper’s ligaments form a distributed fibrous network that is difficult to segment and parameterise patient-specifically. The current model therefore used simplified restoring support contributions to test the effect of additional anterior support, not to reconstruct the true ligament anatomy. Future work could improve this route through distributed support fields, fibre-reinforced material formulations, image-based ligament approximations or better anatomical placement of support regions. Such developments should be validated against surface-motion data before Cooper-like support is used for strong physical conclusions.

5.6 Tumor-overlay route and future tumor sensitivity

The tumor-overlay route successfully added a build-level mechanism for placing analytic spherical stiffness inclusions inside the existing breast volume. This is a useful first step because it allows tumor size and location to be varied without requiring separate Boolean geometry operations for each case. The current

implementation is therefore suitable for preparing systematic tumor-sensitivity studies.

The current tumor representation is still highly simplified. Real breast tumors can differ in size, shape, margin irregularity, stiffness, heterogeneity, depth, local attachment and surrounding tissue context. A spherical homogeneous overlay can test the effect of a controlled stiffness contrast, but it cannot represent the full range of malignant tumor morphology. This limitation is especially important for EWS, because a tumor would only be detectable through its effect on the externally observable surface motion. The relevant question is therefore not only whether a stiff inclusion changes the internal mechanical field, but whether it changes the surface response enough to be measured robustly.

Future tumor studies should systematically vary tumor shape, diameter, stiffness contrast, depth and anatomical location. Central, subareolar, upper-outer, posterior and surface-proximal locations should be compared, and the interaction with fibroglandular tissue and skin should be included. These studies should be run as complete dynamic simulations with tumor-mask post-processing, surface-motion metrics and matched no-tumor controls. Only after such a solved and validated case set is available can the tumor-overlay route be used to discuss potential EWS detectability.

5.7 Computational limitations and solver strategy

A major limitation of the current project was the available solver environment. All COMSOL model builds and time-dependent solves were run on a local laptop/workstation setup. This was sufficient for developing the pipeline, testing individual model routes and generating selected solved cases, but it limited the size and completeness of the final simulation set. Time-dependent FEM simulations with realistic glandular structure, volumetric skin, Cooper-like support, tumor overlays and parameter sweeps are computationally expensive, especially when solver stability and post-processing completeness are required.

Access to faster external computational resources or high-performance computing would substantially strengthen the study. It would make it possible to run larger material-parameter sweeps, complete dynamic tumor-overlay simulations, more robust Cooper-support comparisons, mesh-sensitivity checks, longer simulation windows and repeated simulations for sensitivity analysis. It would also allow more systematic comparison of dynamic loading routes and solver settings. The current computational limitation should therefore be understood as a limitation of project resources rather than as a fundamental limitation of the modelling framework.

Future work should include a clear solver strategy before expanding the model further. This should include mesh-convergence checks, consistent time-stepping settings, automated post-processing for all result groups and predefined criteria for accepting or excluding solved cases. Such a strategy would reduce the risk that model comparisons are influenced by inconsistent solver completion or incomplete output extraction.

5.8 Reproducibility and data management

The project repository supported reproducibility by storing the Python COMSOL pipeline, TOML case definitions, post-processing scripts, compact outputs and report notes in a traceable structure [25]. This was important because the project involved many related model variants, including material, skin, Cooper-support, loading and tumor-overlay routes. Scripted case definitions made it possible to document which assumptions belonged to each variant and reduced the risk of manually losing track of model changes.

At the same time, full computational reproducibility depends on the COMSOL version, license environment and available hardware. Large COMSOL models and generated intermediate files are not always practical to store completely in a report repository. For future work, the most important reproducibility improvement would be to preserve the exact case definitions, solver settings, post-processing scripts and compact numerical outputs for every final result case. This would allow the study to be repeated or extended on a stronger computational system without relying on manual reconstruction of model settings.

5.9 Overall perspective and future work

Overall, this project produced a more anatomically detailed and reproducible COMSOL FEM framework for studying EWS-relevant breast deformation. The model now includes the main components needed for future surface-motion studies: realistic glandular structure, posterior support geometry, material sensitivity routes, volumetric skin, simplified Cooper-like support, dynamic loading variants and analytic tumor-overlay

placement. The current work should be viewed as a model-development step rather than a completed validation study.

The most important next step is to run systematic, fully solved dynamic simulations on stronger computational hardware and compare the simulated surface response with experimental or EWS-like measurement data. This should be combined with controlled variation of material parameters, glandular anatomy, skin properties, loading route and tumor properties. If those future simulations show that tumor-related stiffness contrasts produce measurable and repeatable differences in breast-surface motion, the model could become a useful tool for exploring EWS sensitivity and design choices. Until then, the main conclusion is that the project established a technically traceable and anatomically richer FEM platform for future deformation-based breast-imaging research.

6 Conclusion

References

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Appendix

A Previous EWS FEM work

The present project was developed in the context of previous internal Early Warning Scan (EWS) finite element modelling work. Earlier EWS models provided a methodological starting point for parametrised breast geometry, soft-tissue material assumptions, dynamic loading and displacement-based evaluation [23, 24]. These previous projects were used as internal engineering reference material and as a comparison point for modelling choices. They were not used as validation datasets for the current COMSOL model.

The earlier FEBio-oriented modelling route used a simplified parametrised breast representation in which adipose and glandular regions were defined from geometric construction parameters. Figure 10 shows a schematic cross-section from the earlier EWS project by R. Engels [23]. This representation provided useful context for the transition toward a COMSOL-based workflow, but the current COMSOL model does not reproduce the previous geometry directly.

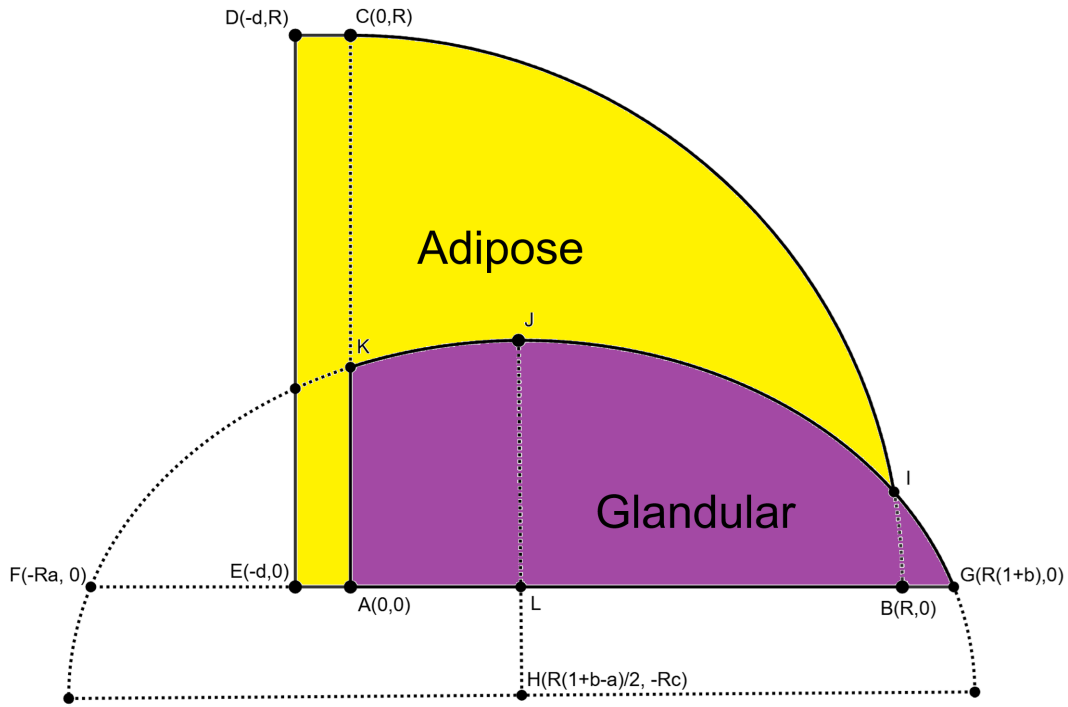


Figure 10: Schematic cross-section from the earlier EWS FEM breast model by R. Engels [23]. The figure illustrates the parametrised adipose and glandular regions used in that previous modelling route and is included here as internal methodological context.

The later EWS/SIOUX workflow provided additional context for skin representation, soft material settings and distributed reinforcement concepts for Cooper-ligament-like support [24]. In the current project, these earlier modelling choices were treated as reference points rather than as fixed requirements. The COMSOL route was developed as a separate implementation with emphasis on controlled geometry construction, automated region definitions, build-only checks and systematic comparison between model variants. This made it possible to investigate selected anatomical and mechanical modelling assumptions in a more traceable way, without implying that the current model is a validated patient-specific representation.

B Current COMSOL repository and reproducibility structure

The active implementation used in this internship project is the COMSOL source-case pipeline developed during this project. The public project repository stores the reproducible parts of the work, including the Python source code, active TOML case definitions, report notes, plotting scripts, compact post-processing tables and selected figures [25]. Large generated COMSOL files, build folders, solve folders, recovery files and raw exports are not intended to be stored in the repository because they can become several gigabytes per case.

The main repository folders used for the current COMSOL workflow are summarised in Table 10. The exact folder structure may still change during repository cleanup, but the distinction between source definitions, generated artefacts and compact evaluation output was maintained throughout the project.

Table 10: Main repository components used in the COMSOL model-development workflow.

Repository component	Role in the project
<code>src/ews_fem_pipeline_comsol</code>	Active Python package for the COMSOL source-case workflow, including case loading, model generation support, Java builder generation, run control and post-processing utilities.
<code>runs/comsol_runs</code>	Version-controlled TOML case definitions for the main model variants and sensitivity studies.
<code>analysis_output/comsol_pipeline</code>	Compact evaluation outputs used for comparison, including summary tables, selected plots and post-processing results.
<code>docs/report_notes/comsol_pipeline</code>	Project notes documenting modelling choices, run status, figure sources and interpretation decisions.
<code>docs/Internship_report_Daan_Kuijpers</code>	The final intership project report.

The COMSOL simulations were run using a licensed COMSOL Multiphysics installation available through TU/e. Most model builds and dynamic solves were executed on local hardware during model development. Because time-dependent COMSOL solves can be computationally expensive, especially for second-order elements, volumetric skin layers and larger parameter sweeps, access to stronger computational resources such as TU/e high-performance computing facilities is relevant for future large-scale studies. In this project, the repository was therefore organised to preserve the reproducible case definitions and compact outputs, while allowing heavy generated artefacts to remain outside version control.

C Simplified example of a COMSOL case definition

The COMSOL pipeline uses TOML case definitions to specify the main model settings. The full case files contain more detailed options than shown below. Listing 1 gives a simplified example to illustrate how geometry, material, loading and evaluation settings are defined in a traceable format.

Listing 1: Simplified example of a COMSOL case definition. Only the main settings are shown; the complete case definitions are stored in the project repository.

```
1 [case]
2 name = "example_reference_case"
3 description = "Simplified reference case for documentation"
4
5 [geometry]
6 breast_volume_ml = 585
7 chestwall_variant = "xoffset055"
8 glandular_distribution = "realistic_reference"
9 outer_shape_asymmetry = "none"
10
11 [materials]
12 adipose_set = "reference"
13 glandular_set = "reference"
14 skin_layer = "none"
15 cooper_support = "none"
16
17 [loading]
18 type = "fixed_support_acceleration"
19 amplitude_g = 0.25
20 duration_s = 0.60
21 mass_damping_alpha = 60
22
23 [tumor]
24 enabled = false
25
26 [evaluation]
27 export_displacement = true
28 export_von_mises_stress = true
29 export_surface_metrics = true
```

The purpose of these case definitions is not only to automate COMSOL runs, but also to make the modelling assumptions explicit. When a model variant is changed, the corresponding TOML file records which geometry, material, loading or evaluation setting was modified. This makes comparisons between model variants more transparent than manually editing a single COMSOL model file for each simulation.

D Breast-volume reference data

An internal Catharina Hospital breast-volume dataset was reviewed to contextualise the selected COMSOL reference volume. The dataset consisted of 100 manually delineated breast volumes. The contours were drawn approximately 5 mm inside the outer skin boundary on images acquired in a supine position. The reported values should therefore be interpreted as internal reference volumes for model scaling, rather than as exact external breast volumes.

Table 11: Summary of the internal Catharina Hospital breast-volume reference dataset. The dataset contained 100 manually delineated breast volumes, with contours drawn approximately 5 mm inside the outer skin boundary in supine images.

Metric	Breast volume (mL)
Mean	728.32
Median	691.15
Standard deviation	327.39
25th percentile	519.65
75th percentile	928.70
Minimum	105.68
Maximum	1777.28

The selected COMSOL reference volume of ~ 585 mL lies between the 25th percentile and the median of this internal dataset. This supports its use as a lower-mid reference volume for the current model-development route. Direct comparison remains limited by the segmentation definition and patient position, because the internal contours were drawn inside the skin boundary and in a supine position.

The internal dataset was also compared with literature values for breast-volume measurement. Stahl et al. reported MRI-based breast anthropometry in a cohort of 400 women and showed a broad distribution of breast volumes [26]. Yoo et al. compared MRI-estimated breast volume with mastectomy specimen weight and supports the clinical relevance of MRI-based volume estimation, although in a cancer/mastectomy cohort [27]. Nie et al. showed that skin removal can change quantitative MRI-based breast-volume and density measurements, which is relevant because the internal dataset was delineated inside the outer skin boundary [28]. Choppin et al. reviewed breast-volume measurement methods and emphasised that measurement uncertainty depends strongly on method, patient pose and boundary definition [29]. Together, these sources support using the ~ 585 mL COMSOL volume as a defensible controlled reference volume, while avoiding overinterpretation as an exact population average.

E Fibroglandular fraction reference data

The selected Stage 3 reference model was compared with literature on volumetric breast density and fibroglandular tissue quantification. This comparison was used to contextualise the chosen fibroglandular fraction, not to assign a clinical BI-RADS category to the model.

Table 12: Literature context used to interpret the selected fibroglandular fraction. The model fraction refers to three-dimensional fibroglandular volume divided by total breast volume.

Source	Relevance for this model
Sartor et al. [30]	Compares qualitative radiologist density classification with automated volumetric assessment and provides context for volumetric density groups.
Nie et al. [31]	Quantifies three-dimensional fibroglandular tissue patterns from MRI and reports volumetric breast and fibroglandular tissue measurements.
Lu et al. [32]	Compares breast tissue measurements across MRI, mammography and mathematical estimation, and shows that individual fibroglandular fractions can vary substantially.
ACR BI-RADS [33]	Provides the clinical breast-composition framework, but these categories are not directly equivalent to the model’s three-dimensional fibroglandular volume fraction.

This density context is also relevant for the EWS application. Women with dense breasts are more difficult to screen using X-ray mammography because fibroglandular tissue can reduce lesion conspicuity and lower mammographic sensitivity [2, 3]. For this reason, a relatively fibroglandular-rich reference model is useful for EWS-oriented model development, provided that it is not interpreted as a population-average breast composition.

The selected COMSOL reference case produced a breast volume of approximately 585 mL and a glandular fraction of 24.308%, corresponding to an approximate glandular volume of 142.2 mL. This value is higher than typical average volumetric density values reported in screening populations, but remains defensible as a controlled fibroglandular-rich reference for EWS-oriented model development. It should therefore be interpreted as a selected anatomical reference configuration rather than as a population-average breast-density value or a direct BI-RADS category.

F Material-parameter literature summary

The material parameters used in the COMSOL model were selected as controlled reference and sensitivity values rather than as patient-specific tissue measurements. Reported breast-tissue stiffness values vary strongly between studies because they depend on measurement protocol, tissue preparation, preload, strain level, loading direction, subject characteristics and constitutive model [4–6,21]. The literature was therefore used to define plausible stiffness scales and sensitivity ranges, not to identify one unique material parameter set.

Table 13: Summary of the literature-based rationale used for selecting the material-parameter scales in the COMSOL model.

Model component	Literature interpretation	Use in the COMSOL model
Adipose tissue	Normal breast adipose tissue is generally reported in the low-kPa stiffness range, although values differ between ex-vivo testing, elastography and model calibration studies [4, 5, 21].	The reference adipose material was assigned an approximate small-strain stiffness of $E \approx 3.66$ kPa. Softer and intermediate variants were included as material-stiffness sensitivity cases.
Fibroglandular tissue	Fibroglandular tissue is typically stiffer and more heterogeneous than adipose tissue. Three-dimensional testing also indicates that human breast tissue can show direction-dependent and spatially heterogeneous behaviour [6].	The reference fibroglandular material was assigned an approximate small-strain stiffness of $E \approx 10$ kPa. Additional variants were used to test the sensitivity of the simulated response to internal tissue stiffness.
Skin layer	Skin is mechanically relevant in breast deformation models and is commonly represented as a stiffer component than the internal breast tissues. Its effective contribution depends on both material stiffness and layer thickness [21, 36].	The active volumetric-skin route used a thickness of 1.5 mm and an approximate stiffness scale of $E \approx 500$ kPa. Lower skin-stiffness cases were retained only as sensitivity variants.
Tumor overlay	Tumor tissue is generally reported to be stiffer than surrounding normal tissue, but the reported stiffness range depends strongly on imaging modality, tissue type and region of interest [4, 7, 8, 39].	The hard tumor-overlay setting targeted an approximate stiffness scale of $E \approx 100$ kPa. This value was used to create a clear stiffness contrast, not as a patient-specific tumor calibration.
Posterior support	The posterior thoracic region was not modelled as a detailed anatomical chest wall. Instead, it was included as a simplified mechanical support region for the breast model.	The constructed posterior support domain was assigned a linear elastic surrogate material with $E = 10$ kPa and $\nu = 0.49$.

Overall, the selected parameters should be interpreted as a controlled modelling set for comparing material routes and sensitivity cases. The values define consistent stiffness scales for the COMSOL workflow, while acknowledging that breast tissue, skin and tumor properties cannot be represented by a single universally accepted parameter value.

G Figure and metric sources

The figures and quantitative values in the main report are based on selected COMSOL outputs, compact post-processing files and generated plots. The complete figure and metric source index is stored in `docs/report_notes/comsol_pipeline/report_figures_metrics_index.md`. This index records the relevant COMSOL output folders, metric files and source images used to create the report figures.

The appendix figures below are included as provenance material. They document the source contact sheets and geometry screenshots used during model development. They are not all intended to carry the same quantitative interpretation as the main results.

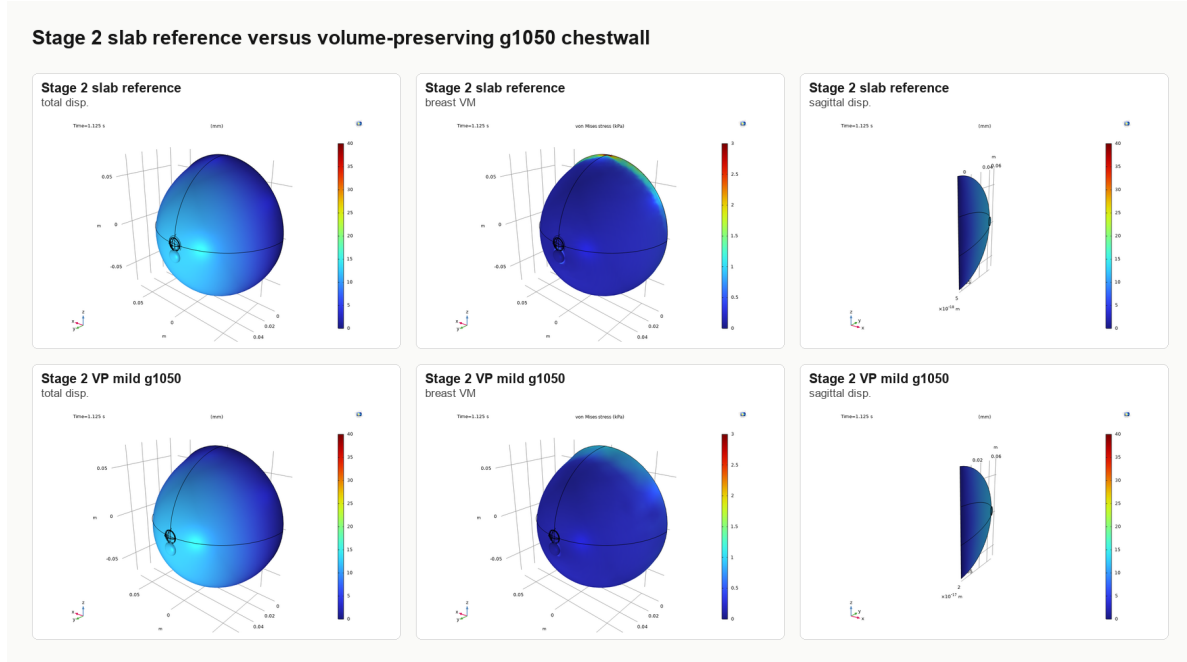


Figure 11: Source contact sheet for the selected chestwall-geometry route. The figure documents displacement and von Mises stress panels used during the evaluation of the curved posterior support geometry.

Stage 3 simple glandular fraction sweep

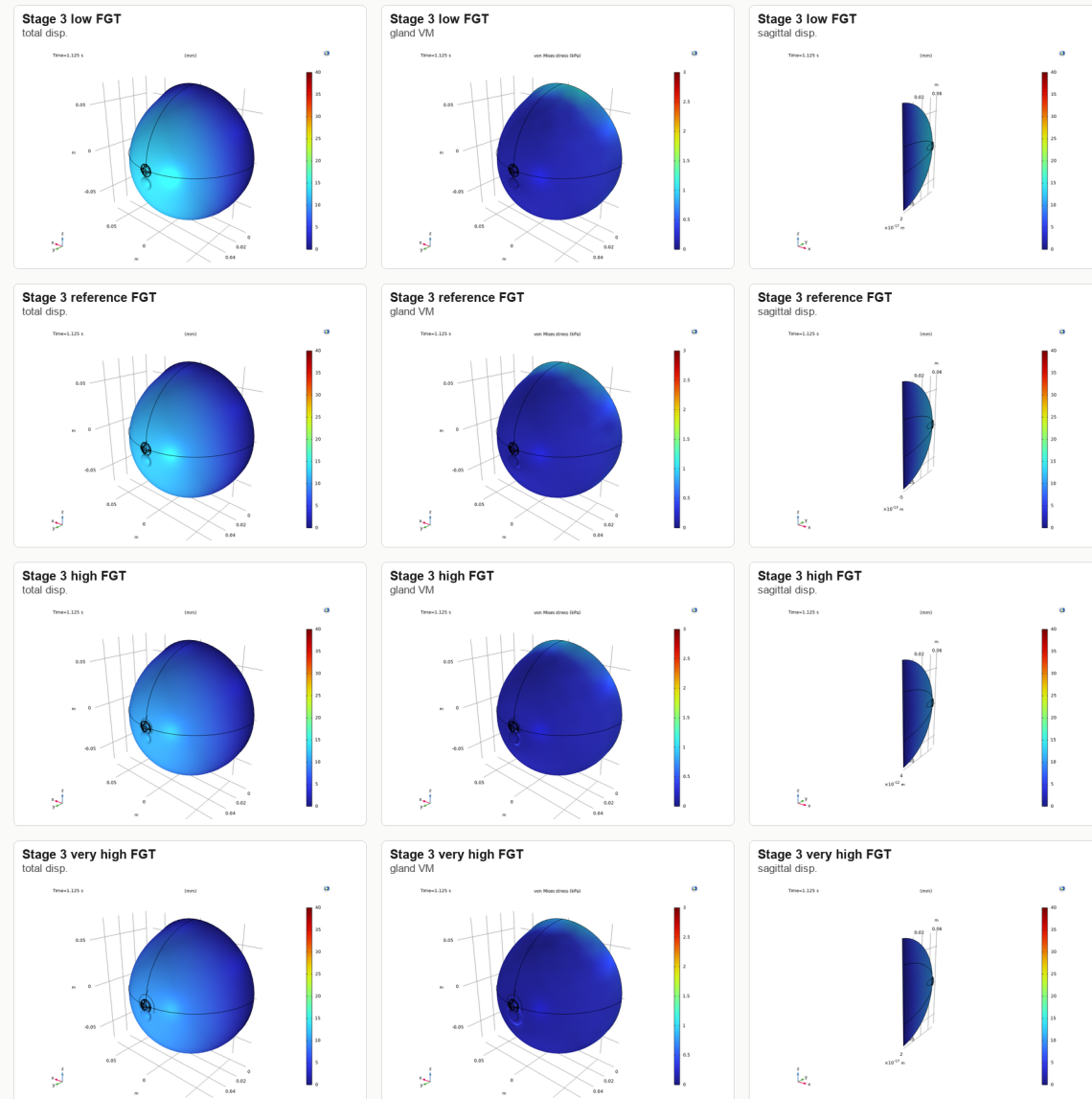


Figure 12: Source contact sheet for the glandular-distribution comparison. The figure documents the solved simple glandular fraction sensitivity and supports the transition toward a more realistic glandular reference.

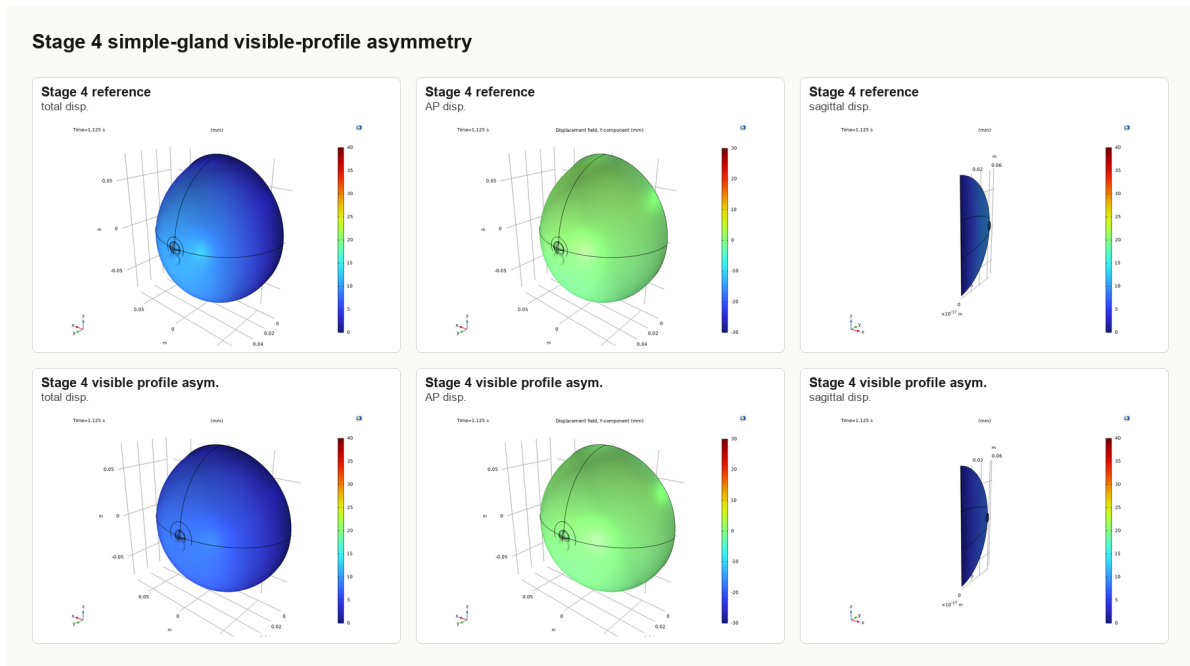


Figure 13: Source contact sheet for outer-shape and nipple-position sensitivity checks. These cases are used as geometry-development context and should be interpreted separately from the matched reference comparisons.

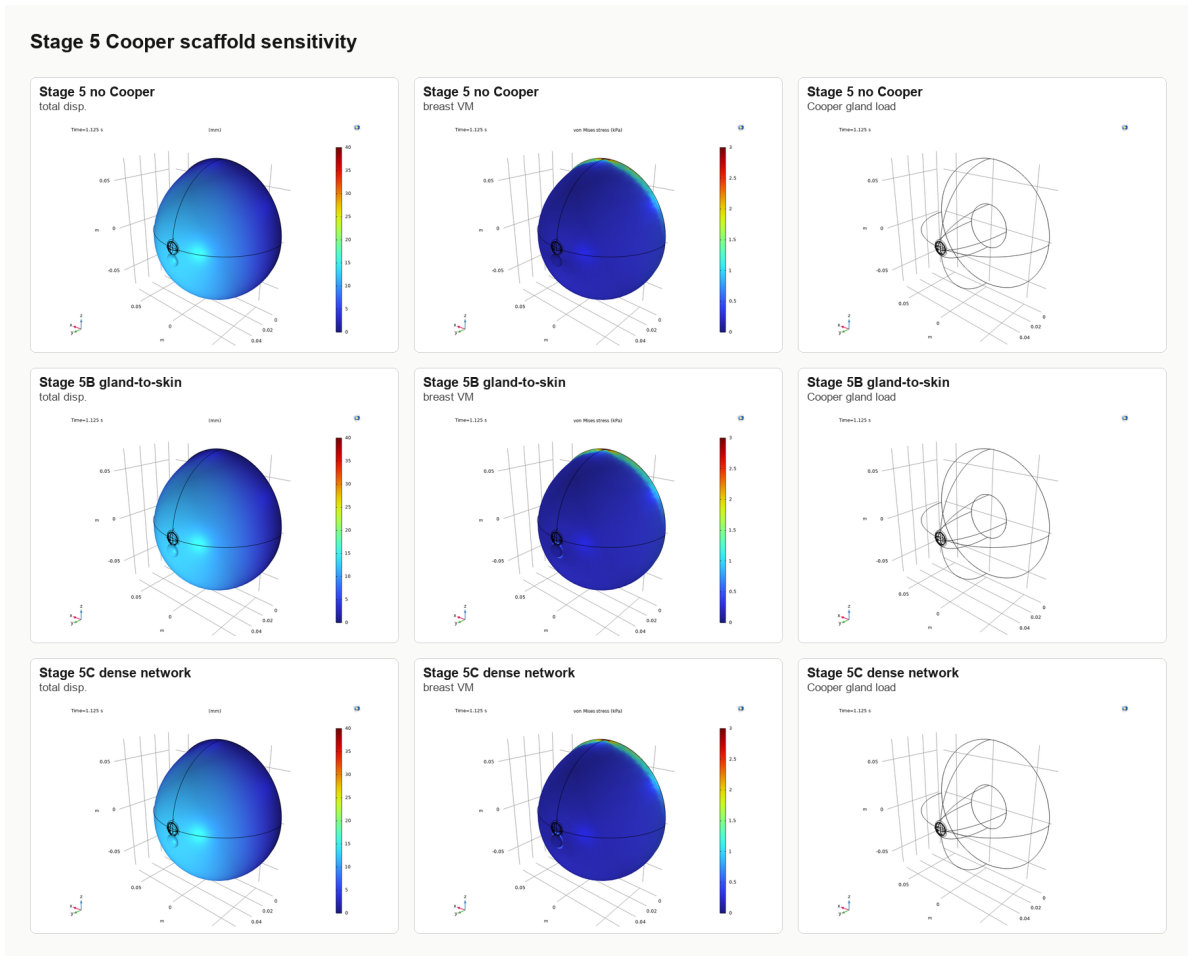


Figure 14: Source contact sheet for the Cooper-support development branch. The stable reference interpretation uses the no-Cooper control unless a Cooper-support variant solves robustly and produces complete post-processing output.